Beyond Stoves and Outdoor Monitors: Indoor PM2.5 Exposure and Inequality in a Fast Growing Megacity

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Exposure to fine particulate matter ($PM_{2.5}$) is the leading environmental risk factor for mortality worldwide. Epidemiological and global health research typically proxies exposure using cooking fuel type and ambient outdoor $PM_{2.5}$ concentrations. Yet urbanization is changing the exposure profile of low- and middle-income country (LMIC) populations who use cleaner fuels and spend most of their time indoors, but live in areas with high outdoor pollution - raising questions about whether either proxy captures actual exposure. We present 152,000 hours of indoor $PM_{2.5}$ monitoring from 300 homes in Jakarta, the world's largest and one of the most polluted cities, where 98% of households have adopted clean cooking fuels. Despite near-universal clean cooking, indoor $PM_{2.5}$ is persistently high — exceeding the WHO daily guideline on 78% of monitored days, with extreme levels invisible to outdoor monitors. We estimate an infiltration factor of 0.62, indicating that the majority of indoor $PM_{2.5}$ originates outdoors. We uncover stark exposure inequality: low-income households experience 113% higher indoor $PM_{2.5}$ than high-income households despite similar outdoor exposure. This inequality is driven by differences in both indoor emission sources, particularly smoking, housing characteristics and ventilation behaviors. Low-cost indoor monitoring reveals what neither cooking fuel type nor outdoor concentrations can capture: actual indoor exposure and its stark distributional consequences in rapidly urbanizing megacities.

Air pollution | Urbanization | Environmental Inequality

Significance Statement

Air pollution kills more people than any other environmental hazard. Yet we have a very limited understanding of what people breathe indoors, which global health research typically infers from two proxies: household cooking fuel type, and outdoor air pollution. Using 152,000 hours of indoor monitoring across 300 homes in Jakarta, the world's largest city, we show that neither proxy works. Despite near-universal adoption of clean cooking fuels, indoor pollution remains persistently high, driven by infiltrated outdoor pollution, smoking, and other indoor sources. Low-income households face more than double the indoor exposure of high-income households, despite similar outdoor pollution levels. Short-term deployment of low-cost indoor sensors uncovers these exposure patterns and inequalities, with direct implications for global air pollution policy.

Fine particulate matter ($PM_{2.5}$) poses the largest environmental threat to human health globally (1). In 2018, among the 7.3 billion people directly exposed to unsafe average annual $PM_{2.5}$ concentrations globally, 80 percent live in low- and middle-income countries (2). Rapidly growing cities face especially elevated levels of ambient air pollution, fueled by intensive use of fossil fuels, industrialization, construction and congestion (3). This pollution burden is expected to intensify: by 2050, an estimated 68% of the global population will reside in cities, with nearly 90% of this growth concentrated in Asia and Africa.

While epidemiological studies and policy evaluations typically rely on ambient $PM_{2.5}$ concentrations as proxies for human exposure, urban dwellers in high-income countries spend approximately 85–90% of their time indoors (4) and likely similar shares in urban LMIC settings. Because indoor concentrations can diverge substantially from ambient outdoor levels depending on building characteristics, ventilation practices, and indoor emission sources, ambient-based exposure assessment may be systematically biased (5, 6). Besides, if the relationship between ambient and indoor $PM_{2.5}$ varies systematically across households—particularly by socioeconomic status, ambient-based exposure assessment may also obscure important dimensions of environmental inequality.

Even though 3.45 billion people in low- and middle-income countries currently live in urban areas (7), we know surprisingly little about their indoor air pollution exposure. Most existing frameworks - including Global Burden of Disease models - emphasize solid cooking fuels as the primary driver of household air pollution (8). This framing is increasingly misaligned with urban realities, where clean cooking fuel adoption has accelerated, but residents remain exposed to severe ambient pollution. A small but growing literature documents high infiltration factors in tropical LMIC cities, but studies typically involve fewer than 50 homes or very limited time coverage (9–13), limiting the ability to identify drivers of heterogeneity. This knowledge gap is fundamentally a data gap. Indoor air pollution data is very scarce, as very few households own indoor monitors, and those that do are generally wealthier and unrepresentative of the broader population.

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We address this critical shortfall by collecting real-time measurements of both indoor and outdoor $PM_{2.5}$ in Jakarta, Indonesia— the world's largest city (14), with over 40 million residents facing severe and persistent air pollution, where traffic, industry, and dense household activity sustain persistently unhealthy air. We equip 308 urban households with indoor air quality monitors, capturing over 152,000 monitor-hours of indoor $PM_{2.5}$ data. Importantly, these monitors did not display $PM_{2.5}$ levels, so the monitors themselves did not induce differential behaviors. We complement these with outdoor $PM_{2.5}$ readings from 63 low-cost sensors, dramatically improving spatial granularity compared to the two reference-grade monitors previously available. Detailed household surveys further document demographic, behavioral, and housing characteristics, making this one of the largest studies on indoor air pollution exposure among urban residents in low- and middle-income countries.

Using this rich dataset, we (1) document indoor $PM_{2.5}$ levels and their relationship to outdoor concentrations, including extreme levels not captured by ambient monitors; (2) estimate infiltration factors, how they depend on housing characteristics and ventilation infrastructure, and compare them to existing LMIC and high-income country estimates; (3) decompose the sources of indoor $PM_{2.5}$, distinguishing outdoor infiltration from hyperlocal sources such as smoking and waste burning; and (4) examine how indoor exposure varies across income levels, despite similar outdoor pollution. Our household surveys provide essential context for interpreting these measurements. Together, these analyses allow us to assess whether ambient $PM_{2.5}$ concentrations adequately capture indoor exposure in a rapidly urbanizing LMIC megacity—and whether the relationship between ambient and indoor pollution differs systematically across socioeconomic groups.

1. Results

A. Households are Exposed to High Indoor $PM_{2.5}$, with Episodic Extremes Missed by Outdoor Monitoring. Our first finding is that average indoor $PM_{2.5}$ concentrations closely map ambient outdoor $PM_{2.5}$ concentrations on average, but episodic spikes of indoor $PM_{2.5}$ are often much higher than those outdoors.Overall, these patterns imply that in Jakarta, households are exposed to higher levels of $PM_{2.5}$ indoors than outdoors on average.

We first establish that both indoor and outdoor $PM_{2.5}$ concentrations are at unhealthy levels. During our study period, the mean indoor daily $PM_{2.5}$ concentration was 40.7 $\mu g/m^3$, while the mean outdoor daily $PM_{2.5}$ concentration was 37.8 $\mu g/m^3$, or an average Indoor/Outdoor (I/O) ratio of 1.08. Both levels were about eight times higher than the WHO annual standard of 5 $\mu g/m^3$, as shown in Fig. 1.

In addition to persistently high pollution levels, we find that indoor spaces also experience extreme episodic events. Hourly indoor pollution levels at the individual household level for a representative two-week period from our sample is displayed in Fig. 1d. The figure shows frequent spikes in indoor pollution, surpassing thresholds typically associated with very unhealthy air quality conditions. For example, while ambient outdoor readings exceeded $100\mu g/m^3$ in 0.7% of the hourly readings, indoor $PM_{2.5}$ readings exceeded this threshold in 7% of the readings.

Given how much time our respondents spend indoors, these measurements are highly representative of personal $PM_{2.5}$ exposure for women and children in our study population. In our accompanying survey, we find that respondents (90% of whom were women) reported spending 80% of their time at home. We focused on parents of elementary school children, who reported that their children spent an average of 70% of their time indoors. Importantly, the sample targets low- and middle-income households, offering rare insights into a population largely overlooked in existing research on indoor air quality. More details on our sampling approach are provided in Section 3.A.

Overall, these findings reveal that while outdoor levels capture average indoor exposure, they fail to account for the pollution burden from episodic indoor spikes. Individuals with similar average exposures may experience very different intensities and frequencies of severe pollution events, patterns that are missed when relying solely on outdoor pollution data. The health impacts of repeated short-term spikes remain poorly understood, underscoring the need for research that considers pollution spikes, not just average levels.

B. High Infiltration of Outdoor $PM_{2.5}$ into Indoor Spaces. To assess the role of ambient outdoor $PM_{2.5}$ in high indoor $PM_{2.5}$ levels, we model the infiltration of the former into indoor spaces, where ambient outdoor refers to background $PM_{2.5}$ concentrations measured by our citywide sensor network.

We estimate the infiltration factor of ambient outdoor $PM_{2.5}$. We find that a 1 $\mu g/m^3$ increase in ambient outdoor $PM_{2.5}$ is associated with a .62 (95% CI, 0.47, 0.77; $P < 0.001$) $\mu g/m^3$ increase in indoor $PM_{2.5}$ in the same hour. This number is 2 to 3 times the infiltration factor typically estimated in the United States (15, 16) and is aligned with factors estimated in low- and middle-income settings across a limited number of houses or time period (17).

These estimates are robust to alternative regression models, as seen in Fig. 2a. We also estimated a distributed lagged model, showing that infiltration is happening within the hour (*SI Appendix*, Fig. S1), as opposed to estimated lagged effects up to 6 hours in high-income settings (15, 16). While many studies use daily or multi-hour averaged concentrations to approximate steady-state conditions and to absorb lagged dynamics between outdoor and indoor $PM_{2.5}$ (17), our distributed lag model confirms that the outdoor-indoor relationship is essentially contemporaneous in our setting, justifying the use of hourly observations. We also note that aggregating to 24-hour means would substantially reduce the effective sample size, given that we were unable to collect continuous data for every single hour for all households (Section A and *SI Appendix*, Fig. S2).

Ambient outdoor $PM_{2.5}$ is measured with error, since sensors cannot capture the hyperlocal conditions at each household. Measurement error would attenuate the estimated infiltration coefficient toward zero, with greater attenuation for households located farther from a sensor. Consistent with this, we show in *SI Appendix*, Fig. S3, that the estimated infiltration factor is stable at approximately 0.80 when we restrict to households within 500m, 1km, and 2km of an outdoor sensor, before declining to 0.62 in the full sample.

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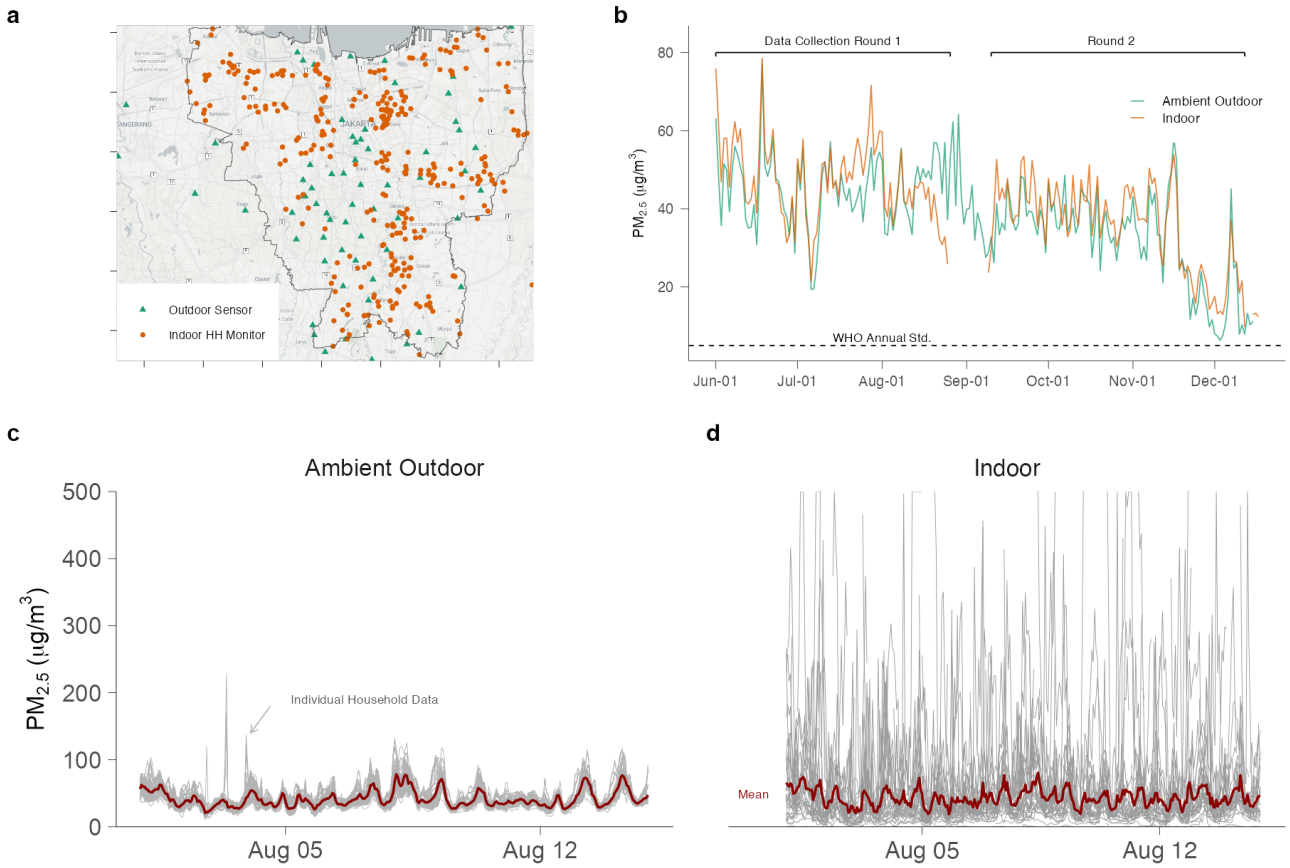
Fig. 1. Indoor and Outdoor $PM_{2.5}$ in Jakarta



(a) Locations of respondents and installed air quality monitoring hardware. There are 308 indoor air quality monitors (orange) installed in respondent homes, and 63 low-cost sensors (green) to measure respondent-specific outdoor air quality. **(b)** Daily average outdoor and indoor pollution levels in Jakarta. To account for the uneven spatial distribution of outdoor sensors, we first computed district-level averages for outdoor readings, which were then aggregated to produce a daily citywide average. Indoor pollution levels were averaged across all households on a daily basis. The gap in the indoor data from August 26 to September 8 occurred during a transition period when the air quality hardware was being moved between households. **(c)** Hourly outdoor $PM_{2.5}$ from August 1–14, 2024. Gray lines show hourly $PM_{2.5}$ concentrations associated with each individual, while the maroon line shows the mean concentration across all 68 sensors. **(d)** Analogous to (c), this plot shows hourly indoor $PM_{2.5}$, with gray lines corresponding to each individual low-cost sensor.

This suggests that our main estimate represents a conservative estimate of the true infiltration factor.*

We leverage the breadth of our survey data to examine drivers of variation in the infiltration factor (Fig. 2b). Our results suggest that differences in infiltration factors are partly driven by long-term household structural characteristics and in particular ventilation, which influences the air exchange rate (17). In particular, we find suggestive differences based on air conditioning (AC) ownership ($p = 0.14$) and whether the monitor room has an open versus closed window or door ($p = 0.15$) during the measurement hour. In Jakarta's hot and humid climate, households without AC often rely on natural ventilation, increasing their exposure to outdoor pollution.

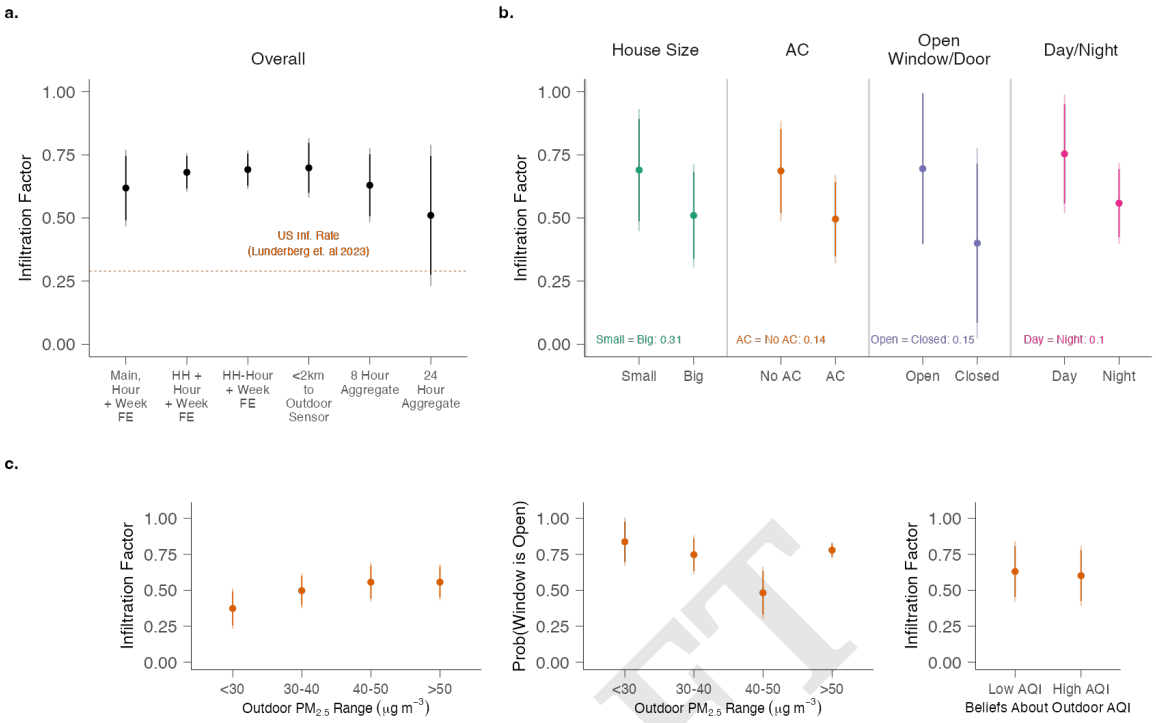
In high-income settings, infiltration factors are primarily driven by short-term behavioral adjustments (15, 18). We find no evidence of such responses in Jakarta. Three analyses show that households are not actively taking short-term actions to reduce infiltration.

First, while previous literature has found that infiltration factors decline at high outdoor $PM_{2.5}$ concentrations, suggesting that people actively modify their behavior to limit exposure at higher levels of outdoor pollution (15, 16), we do not find evidence of this relationship in our setting. In fact, infiltration increases at higher $PM_{2.5}$ concentrations. This pattern, shown in Fig. 2c, could have several drivers, including concentration gradients between outdoor and indoor pollution, meteorological conditions correlated with outdoor pollution levels, or the presence of high-emission hyperlocal pollution sources (16, 19).

Second, we assess whether ventilation behaviors (e.g., opening windows or doors) respond to outdoor pollution levels. If households actively manage infiltration, we would expect them to close windows when pollution is high. However, Fig. 2c shows no clear relationship between window or door openness and outdoor pollution levels. While opening or closing windows is a short-term behavior that can, in theory, be adapted, we may expect that households do not keep

*Pairwise correlations between outdoor sensors average approximately $R^2 = 0.6$ across the city and remain above 0.5 even when sensors are 30km away from each other (SI Appendix, Fig. S1). This indicates that ambient outdoor $PM_{2.5}$ in Jakarta is largely driven by city-wide variation and that readings from individual sensors are reasonably representative of broader conditions. However, the decline in correlation with proximity — from R^2 0.75 for nearby sensors to R^2 0.50 at greater distances — confirms that hyperlocal variation exists, consistent with the attenuation in our estimated infiltration factor for households farther from sensors (*SI Appendix*, Fig. S3).

Fig. 2. Infiltration Factor from Ambient Outdoor to Indoor Homes



(a) Infiltration factors, defined by the regression of indoor $PM_{2.5}$ on outdoor $PM_{2.5}$ (Equation 1). Each point corresponds to a different specification. The first specification controls for hyperlocal source variables (waste burning frequency, household smoking, kitchen source presence, cooking activity, and distance to main road), outdoor temperature, humidity, and hour-of-day and week fixed effects. The second and third add household fixed effects and household-by-hour fixed effects, respectively. The fourth restricts the sample to households within 2 km of an outdoor sensor. The fifth and sixth aggregate observations to 8-hour and 24-hour means, respectively. All regressions control for outdoor temperature and humidity. **(b)** Heterogeneity in infiltration factors by household and behavioral characteristics, including house size, air conditioning (AC) presence, whether a window or door was open, and time of day (day/night). Each regression includes hyperlocal source controls, outdoor temperature, humidity, and hour-of-day and week fixed effects. P-values for tests of equality between coefficients are shown in the respective panels. **(c)** Household behavior in response to outdoor pollution levels. The first panel shows heterogeneity in infiltration factors as a function of outdoor $PM_{2.5}$ concentrations, binned into four ranges. The second panel shows the probability of having a window or door open as a function of outdoor $PM_{2.5}$ concentrations. The third panel shows heterogeneity in infiltration factors based on households' baseline beliefs about whether outdoor air quality is unhealthy (high AQI) versus not (low AQI). Points represent regression coefficients; darker error bars show 90% confidence intervals, and lighter error bars show 95% confidence intervals. Errors are clustered at the household and date-hour level.

DRAFT

windows closed in a hot and humid climate like Jakarta, where natural ventilation is important for cooling.

Finally, we find no evidence that beliefs translate into meaningful protective action. Infiltration factors do not differ between households who perceive outdoor air as unhealthy and those who consider it relatively healthy, as shown in Fig. 2c. Households were classified based on whether they generally thought outdoor air pollution was unhealthy (high AQI vs. low AQI) before monitors were installed. The disconnection between beliefs and protective action is consistent with other research finding low protective behavior in low-income settings (20, 21).

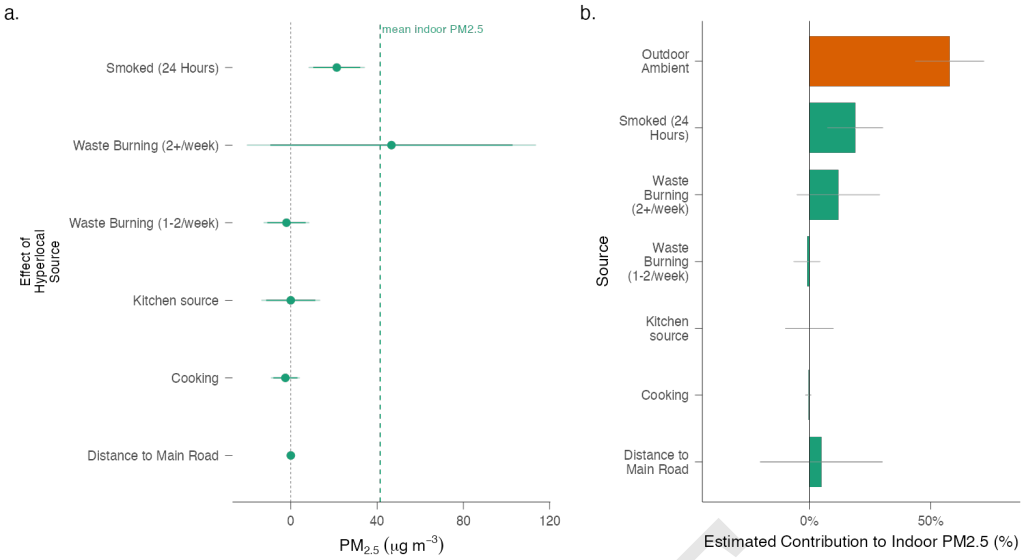
C. Indoor and Hyperlocal Sources of Indoor $PM_{2.5}$. We leverage our rich panel of $PM_{2.5}$ data and survey to quantify the contribution of different sources of indoor $PM_{2.5}$. With these high infiltration factors, most indoor air pollution in Jakarta originates from the outdoors. Overall, we estimate that 58% of mean indoor $PM_{2.5}$ comes from ambient outdoor pollution (Fig. 3b), a larger share than in high-income countries (16, 22).

While ambient outdoor pollution is the primary driver of persistently high indoor $PM_{2.5}$ concentrations, our surveys shed light on the important role of indoor and hyperlocal sources in both average levels and episodic spikes of indoor $PM_{2.5}$. In particular, in Fig. 3, we document the clear impact of two major sources: indoor smoking and neighborhood waste burning.

We find that households who reported indoor smoking within the last 24 hours experienced significantly higher indoor pollution levels. Smoking was associated with 21 $\mu g/m^3$ (95% CI: 8.30–34.3, $P < 0.001$) higher average hourly indoor $PM_{2.5}$ concentrations, a 51% increase compared to the sample mean. These patterns are remarkably well aligned with global evidence that average measured indoor $PM_{2.5}$ is 20 to 30 $\mu g/m^3$ higher in smoking than non-smoking households (23, 24). In our model, smoking explains approximately 19% of the variation in mean indoor $PM_{2.5}$.

Neighborhood-level waste burning is another significant source of indoor air pollution in Jakarta. We estimate that households in areas with more than two waste-burning events per week (11% of the sample) experienced higher average indoor $PM_{2.5}$ concentration; this explains approximately 12% of the variation in mean indoor $PM_{2.5}$. However, accurately quantifying the effect of burning events on indoor pollution is challenging because of our reliance on self-reported data and some events may be misreported. These measurement

Fig. 3. Source Contributions to Indoor $PM_{2.5}$



(a) Effects of hyperlocal pollution sources on indoor $PM_{2.5}$, estimated from Equation 1: whether the household reported any smoking indoors in the 24 hours preceding the interview, whether the household reported observing 1-2 or 3+ waste burning events in the neighborhood in the 7 days preceding the interview, whether the monitor is in or near the kitchen, whether the household was cooking at the time of measurement, and distance to the nearest primary road (km). The model controls for outdoor temperature, and humidity, with hour-of-day and week fixed effects. The dashed green line indicates mean indoor $PM_{2.5}$, providing a benchmark against which to interpret the magnitude of each hyperlocal source coefficient. **(b)** Estimated mean contribution of each source to indoor $PM_{2.5}$, calculated by multiplying each regression coefficient by the sample mean of its corresponding variable and expressing the product as a share of mean indoor $PM_{2.5}$. Error bars are computed by scaling the upper and lower bounds of each coefficient's confidence interval by the sample mean of the corresponding variable and dividing by mean indoor $PM_{2.5}$. Points represent regression coefficients; darker error bars show 90% confidence intervals, and lighter error bars show 95% confidence intervals. Standard errors are clustered at the household and date-hour level.

DRAFT

limitations contribute to wide confidence intervals around our estimates.

Beyond the difficulty of surveying, waste burning occurs outdoors but at spatial scales finer than our sensor network resolves. Thus it is likely that outdoor sensors partially capture these burn events: households reporting more than two burning events per week have outdoor $PM_{2.5}$ that are 3.83 $\mu g/m^3$ higher on average, increasing to 5.49 $\mu g/m^3$ for households living within 2 km of a sensor (*SI Appendix*). This is consistent with evidence that $PM_{2.5}$ from ground-level combustion decays over hundreds of meters (25). However, households reporting frequent burning may also live near other outdoor pollution sources, so we cannot cleanly attribute this differential to waste burning alone. We thus suggest interpreting these estimates with caution.[†]

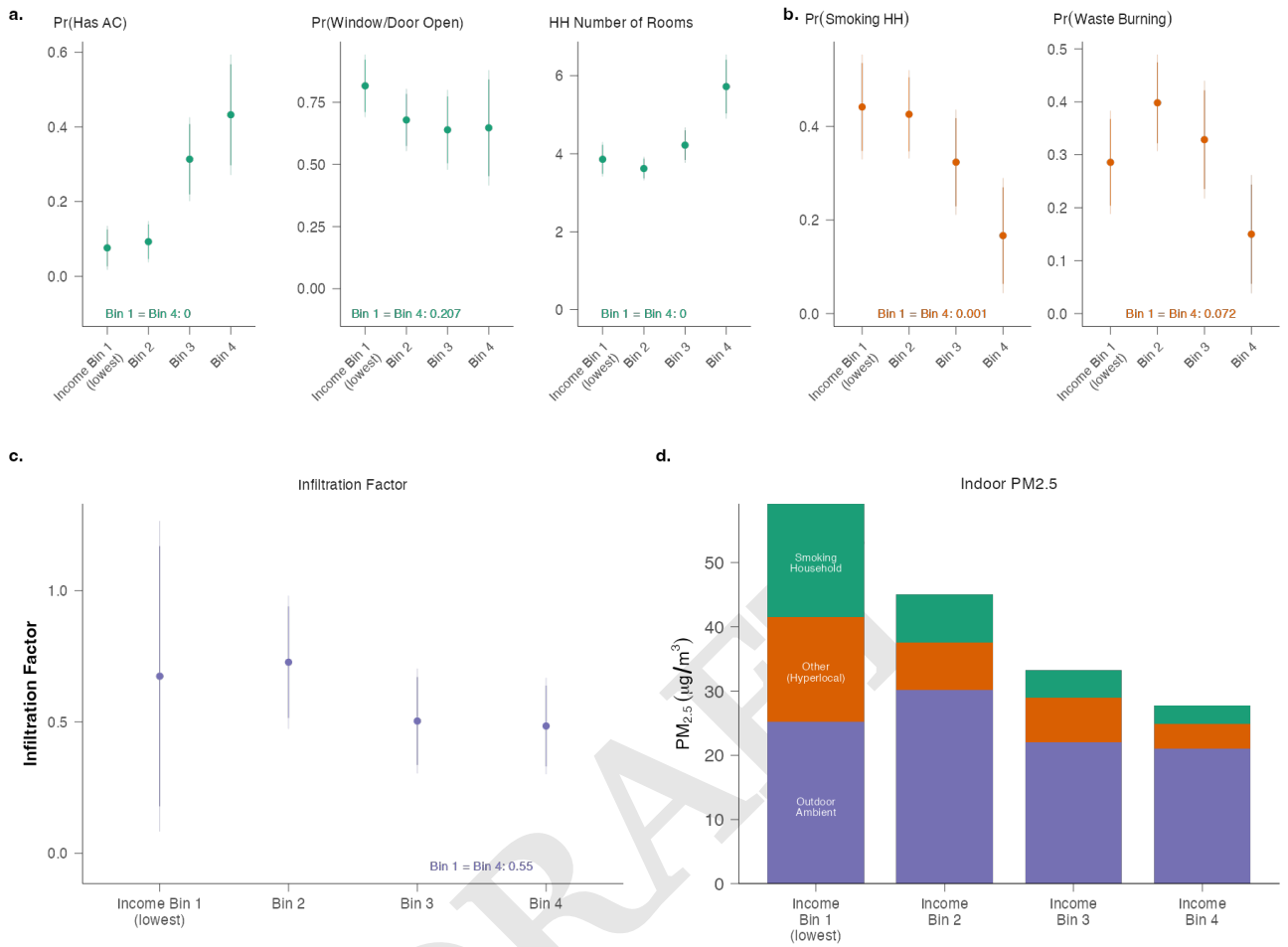
We additionally examine episodic PM2.5 spikes, defined as hours in which the hour-to-hour change in indoor PM2.5 exceeds one standard deviation of the overall indoor $PM_{2.5}$ distribution. Episodic spikes are of particular interest because sharp hourly increases are more likely to reflect discrete hyperlocal emission events than the gradual infiltration of ambient outdoor pollution, which varies more smoothly over time. Notably, the probability of an episodic spike was 341.8% higher in smoking households than in non-smoking households. Smoking is the dominant contributor, accounting for 56% of the variation in indoor $PM_{2.5}$ spikes, followed by outdoor $PM_{2.5}$ (29%) and waste burning (24%). We present a decomposition of source contributions to spikes in *SI Appendix*, Fig. S8, mirroring the analysis shown in Fig. 3, and the analogous LMG R^2 decomposition in *SI Appendix*, Table S3.

In addition to smoking and neighborhood waste burning, there are likely other hyperlocal pollution sources contributing to the observed levels and episodic spikes in indoor air pollution but our context is not suitable to draw any conclusive evidence on these. For example, a major contributor to indoor air pollution that has been documented extensively is the combustion of solid fuels and kerosene for cooking. However, in Indonesia, the widespread adoption of LPG cookstoves has largely eliminated variation in cooking fuel use (26); in our survey, 98% of households reported using LPG cookstoves for cooking. While cooking with LPG cookstoves may still generate large episodic spikes in pollution, our data do not include detailed information on cooking activities, preventing us from drawing insights on this source of indoor pollution. Besides, we do not find that typical cooking hours and whether the air monitor is located in the kitchen impacts measured levels of indoor $PM_{2.5}$ (Fig. 3), which may be due partly to the small size and open layout of housing dwellings in our setting. Analogously, the density of Jakarta's road network may explain the absence of estimated explanatory power of distance to a primary road on indoor $PM_{2.5}$ levels. For example, 58% of our sample lives within 500 meters from a primary road.

D. Heterogeneity by Income. Households across the income distribution experience similar levels of ambient outdoor pollution exposure, but lower-income households are exposed to higher levels of indoor $PM_{2.5}$ (*SI Appendix*, Fig. S6). Moving from the highest to the lowest income quartile in our sample is associated with an increase of 23.7 $\mu g/m^3$

[†] As a complementary measure of source importance, an LMG R^2 decomposition yields a ranking of sources consistent with Fig. 3 (*SI Appendix*, Table S2).

Fig. 4. Indoor $PM_{2.5}$ Higher for Lower Income Households



(a) Housing characteristics by income quartile: probability of having air conditioning, probability of having a window or door open at the time of interview, and number of rooms in the household. Each point is the quartile-specific mean from a regression of the characteristic on income quartile indicators with no intercept. **(b)** Prevalence of hyperlocal pollution sources by income quartile: probability of any household member smoking in the past 24 hours, and probability of observing neighborhood waste burning in the past 7 days. **(c)** Infiltration factors by income quartile, estimated by interacting outdoor $PM_{2.5}$ with income quartile indicators in Equation 1, controlling for hyperlocal source variables, outdoor temperature, humidity, and hour-of-day and week fixed effects. **(d)** Decomposition of mean indoor $PM_{2.5}$ by income quartile into outdoor ambient (purple), other hyperlocal sources (orange), and smoking household (green) contributions. The outdoor ambient contribution is calculated by multiplying income-quartile-specific infiltration factors by quartile-specific mean outdoor $PM_{2.5}$. The smoking contribution is calculated analogously using quartile-specific smoking coefficients and prevalence. The remaining difference is attributed to other hyperlocal sources. P-values for tests of equality between Bin 1 and Bin 4 are shown in each panel. Darker error bars show 90% confidence intervals, and lighter error bars show 95% confidence intervals. Standard errors are clustered at the household and date-hour level.

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in mean $PM_{2.5}$. Overall the lowest-income households experience nearly double the indoor $PM_{2.5}$ levels of the highest-income households. This finding is consistent across various aggregation techniques used to calculate the mean (*SI Appendix*, Fig. S6).

We find that this heterogeneity in indoor $PM_{2.5}$ exposure across the income spectrum stems from both housing and ventilation characteristics, and the prevalence of indoor sources.

Income may affect infiltration of outdoor pollution into indoor spaces through its relationship with housing quality and ventilation infrastructure (16). In Fig. 4c, we show suggestive evidence that the estimated infiltration factor is higher for the lowest income quartile than for the highest income quartile. Estimated infiltration factors are 0.69 (Bin 1) and 0.48 (Bin 4); although Bin 1 is higher, the Bin 1 vs Bin 4 difference is statistically noisy ($p = 0.53$). Overall, we estimate that differences in infiltration explain 16% of the difference in average indoor $PM_{2.5}$ between the lowest and highest income quartiles.

Although our design does not permit causal identification of specific mechanisms, survey responses suggest several contributing factors for heterogeneity in infiltration rates. As shown in Fig. 4a, higher-income households are significantly more likely to have air conditioning, reducing their reliance on open windows or doors for ventilation. Consistent with this pattern, households in the lowest income quartile are more likely to keep windows or doors open in the air quality monitor room at the time of surveying, although this difference is not statistically significant as shown in Fig. 4a. Higher-income households also tend to live in larger homes, both decreasing the concentration of pollutants and potentially enabling them to separate living areas from cooking and smoking areas, for example. Taken together, these patterns are

consistent with lower-income households having lower housing quality and facing greater challenges in regulating indoor temperatures, making them more vulnerable to outdoor pollution infiltration, as discussed in Section B.

Turning to hyperlocal sources, we find that smoking is the biggest driver of heterogeneity between income groups. In Fig. 4b, we show that indoor smoking is significantly less common among higher-income households, translating into a lower contribution of smoking to indoor $PM_{2.5}$ as shown in Fig. 4d. Overall, we estimate that smoking differences explain 48% of the difference in average indoor $PM_{2.5}$ between the lowest and highest income quartiles.

We estimate that differences in other hyperlocal pollution sources explain the remaining 38% of the difference in average indoor $PM_{2.5}$ between the lowest and highest income quartile. Consistently with this pattern, we find that waste burning is about half as likely to be reported by participants in the highest income quartile than other participants, though measurement challenges limit the robustness of this finding. While cooking fuel combustion has been documented as another major contributor to indoor air pollution, the near-universal adoption of LPG cookstoves in our sample largely eliminates variation in cooking fuel use, limiting our ability to further investigate this channel.

The heterogeneity of indoor and hyperlocal pollution sources across the income gradient is also consistent with a greater burden of episodic pollution for low-income households. We find that households in the lowest income quartile experience unhealthy pollution spikes 3% of the time, compared to just 0.4% for households in the highest income quartile, a statistically significant difference ($p = 0.027$). Since ambient outdoor pollution rarely reaches such extreme levels in Jakarta, these spikes likely result from more frequent episodic emissions occurring inside or near the home, which may be affected by behaviors.

2. Discussion

We show that indoor $PM_{2.5}$ exposure in the world's most populated city is persistently high, exceeding the WHO daily guideline on 78% of monitored days. While high indoor levels are primarily driven by infiltration from ambient outdoor $PM_{2.5}$, outdoor concentrations are an imperfect proxy for exposure. Indoor and hyperlocal sources contribute significantly to pollution levels and induce hazardous spikes, extremes not captured by outdoor monitors. Despite comparable ambient outdoor PM2.5 levels, households in the lowest income quartile experience nearly twice the indoor PM2.5 concentration of those in the highest quartile. This disparity is largely driven by smoking and other indoor sources, with additional evidence suggesting that differences in infiltration rates also play a role. Neither cooking fuel type nor outdoor concentrations, the proxies commonly used in global health frameworks and epidemiological studies, fully capture these patterns.

We estimate an infiltration factor between 0.51 and 0.71. This estimate is broadly consistent with, and extend, a growing body of evidence from urban LMIC settings such as Dhaka (9, 10), Mumbai (11), Hong Kong (12) and Kampala (13). This evidence has, to date, been largely constrained by small sample sizes (typically 6–43) and short monitoring windows (typically 24 hours). With over 152,000 monitor-hours across more than 300 homes, together with detailed household surveys, our study offers substantially greater statistical power to characterize the distribution of indoor $PM_{2.5}$, the dynamic of infiltration of outdoor air into indoor spaces, how infiltration depends on housing characteristics and ventilation infrastructure, the contribution of indoor and hyperlocal behaviors on average daily pollution and pollution spikes, as well as their determinants across a socioeconomically diverse urban population. Our estimates are significantly different than for high-income settings (15, 16, 27), emphasizing the need for more research in LMIC cities.

We find that most indoor pollution stems from outdoor $PM_{2.5}$ infiltration, with an infiltration factor (0.62) substantially higher than estimates in high-income countries. This high infiltration factor is likely driven by both a high penetration factor and a high air exchange rate, reflecting the prevalence of natural ventilation and Jakarta's hot and humid climate. Unlike in high-income settings where households reduce infiltration during high pollution episodes by closing windows and using air filtration (15), we find no evidence of such protective behavioral responses in Jakarta. This suggests that avoidance strategies effective in HICs may not transfer to LMIC contexts, where air conditioning is less prevalent and heat limits the ability to keep windows closed.

A simple back of the envelope calculation (*SI Appendix*, Table S1) confirms that infrastructure improvements like weatherization (for example sealing cracks and gaps in the walls) could reduce indoor $PM_{2.5}$ concentrations, although a full cost-benefit comparison between outdoor source reduction and weatherization is beyond the scope of this study. However, closing windows does not unambiguously reduce indoor $PM_{2.5}$. For households with significant indoor sources — particularly smokers — reduced ventilation limits dilution of internally generated pollution, potentially raising concentrations above what they would have been with windows open. Stronger enforcement against garbage burning or information campaigns against indoor smoking (28) could not only reduce these extreme events and their associated health risks but also increase the relevance of changing ventilation behaviors. The contribution of these indoor sources also raises mismeasurement concerns when ambient outdoor $PM_{2.5}$ is used as a proxy for exposure even though indoor-sourced episodic peaks are not captured by outdoor monitors.

Our findings have important distributional implications. Consistent with previous studies (15, 20, 29), we show that low-income households are exposed to the highest levels of indoor air pollution. The socioeconomic gradient in indoor $PM_{2.5}$ concentration is not due to heterogeneous ambient outdoor $PM_{2.5}$ but rather to higher prevalence of indoor and hyperlocal sources in lower income households, as well as lower prevalence of air conditioning. Accurate exposure assessment may therefore require coupling outdoor air quality mapping with demographically representative indoor measurements.

Importantly, we estimated this large contribution of indoor and hyperlocal sources to indoor $PM_{2.5}$ concentrations despite 98% of households in our sample using LPG cookstoves—a fuel typically classified as clean cooking. This highlights the need for a broader framework for understanding indoor air pollution in low- and middle-income countries. For example, the Global Burden of Disease (GBD) framework

DRAFT

defines household air pollution in terms of solid fuel use—a metric that has guided policy and resource allocation globally. However, this definition does not fully capture the indoor pollution risks experienced by urban populations that use cleaner fuels but remain exposed to high ambient and infiltrated pollution, as well as additional indoor or hyperlocal sources like smoking or trash burning.

Despite these contributions, our study has several limitations. First, our sample focuses on households with elementary school-aged children, and therefore is not fully representative of Jakarta's population. Future research should assess whether similar patterns hold for other demographic groups, particularly groups with heightened health risks, such as the elderly or those with pre-existing health conditions. Second, a formal co-location study of the sensors used in our indoor monitors against a reference-grade monitor in Jakarta would strengthen future work in this setting. Third, while we identify key determinants of indoor air quality, we do not capture all possible hyperlocal pollution sources, such as cooking emissions, incense burning, or traffic-related infiltration, nor are we able to track personal exposure outside the home. Additionally, while our estimates suggest that infiltration factors vary by income, we lack the statistical power to establish this precisely. Larger samples could clarify whether housing characteristics like air conditioning and ventilation infrastructure drive systematic differences in infiltration across income groups. Relatedly, our reliance on stationary indoor monitors limits our ability to account for variations in individual exposure as people move between environments. Expanding real-time monitoring to include wearable sensors and more detailed measurements of indoor pollution sources could provide a fuller picture of exposure risks.

3. Materials and Methods

High levels of outdoor air pollution are sustained in Jakarta throughout the whole year, and historically especially high between the months of May and October. We designed the study to cover this high pollution season and conducted it in two successive rounds, with approximately 154 households participating in each round. Round 1 was conducted from 1 June to 28 August 2024, and Round 2 was conducted from 10 September to 10 December 2024. This randomized controlled trial was registered on the AEA RCT registry (Identification No. AEARCTR-0013110) and received IRB approval from the University of Chicago (Protocol No. IRB23-0310) and from Badan Riset dan Inovasi Nasional (Application No. 29012024000032). Individual respondents consented to the collection of the indoor air quality data and survey data used in this study. Ambient outdoor $PM_{2.5}$ data comes from devices monitored by our hardware implementation partner, not associated with private individuals, and therefore without personally identifying information.

A. Study Participants and Datasets.

Study Participants: Households participating in our study were selected among Jakarta households, through a partnership with the Education Office in Jakarta, which compiled a representative list of Jakarta schools. We randomly sampled households from school rosters and successfully enrolled 308 participants, 285 of whom successfully installed the hardware, connected to Wi-Fi, and began data collection. Therefore, all our participant households have at least one elementary school child. As surveying challenges like low participation and high attrition are common in developing country cities (20), our sample strategy was designed to address these challenges and ensure the proper use and return of air quality hardware. By the end of our study, attrition was only 6%. This attrition rate is consistent with rates typically observed in rural contexts, but is an exceptionally low rate for surveying in urban settings. For instance, (20), a paper with a similar research question conducted in Delhi, reported attrition rates of 19–32% across their treatment groups. All analyses were conducted in R version 4.3.2.

The geographic distribution of participating households is displayed in Fig. 1a, which demonstrates the wide coverage across the city. Households in our sample are relatively larger and less wealthy than the average Jakarta household but span a wide range of socioeconomic backgrounds. For more information on how our sample compares to the wider Jakarta population, see *SI Appendix*, Table S3.

Following enrollment, the survey enumerators visited households to obtain informed consent and administer a baseline survey. This survey captured household demographic information, housing structure characteristics, and beliefs and behaviors related to indoor and outdoor air pollution. Upon consent, air quality hardware was installed in the home for approximately three months, during which it collected continuous indoor air quality data. Survey questions are listed in the *SI Appendix*.

Indoor $PM_{2.5}$: The air quality hardware consisted of a fan equipped with an internal $PM_{2.5}$ sensor (Cubic PM2008M) and was installed in each participating household. The sensor measures particulate concentration via an optical light-scattering method, with accuracy specifications based on factory calibration against a GRIMM aerosol spectrometer. The sensors were designed to collect $PM_{2.5}$ data in real time, with readings not visible to respondents. When connected to Wi-Fi, the devices transmitted data to a remote server at five-minute intervals, which were subsequently aggregated to the hourly level for analysis. The hardware was produced by Nafas, a Jakarta-based air quality technology company.

To incentivize households to keep the hardware on, participants received IDR 35,000 (approximately USD 2) every 30 days, contingent on the hardware being connected to Wi-Fi and operational for an average of 14 hours per day. For households without Wi-Fi or unwilling to use their Wi-Fi for the experiment, dedicated Wi-Fi units were installed. To offset electricity costs, participants received an additional IDR 15,000 (approximately USD 1) every 30 days.

Wi-Fi connectivity issues led to intermittent data loss; in these cases, enumerators were dispatched to retrieve readings manually. Data coverage averaged 1,138 hours per household over the 2–3 month monitoring period. However, coverage varied considerably across households: while most contributed sufficient data for analysis, a subset experienced persistent Wi-Fi signal failure that limited retrievable readings to only a few hours. The full distribution of household-level coverage is reported in *SI Appendix*, Fig. S2.

To assess whether non-random patterns of missing data affect our estimates, we re-estimate Equation 1 under three alternative inverse-probability weighting schemes. Our main

DRAFT

specification does not include weights, which implicitly upweights households for whom receive more $PM_{2.5}$ readings. As robustness checks, we alternatively weight by the inverse of (i) the household-specific observation rate, upweighting households with more missing data, and (ii) the household-by-hour observation rate, upweighting household-hour cells with more missing data. If missingness were systematically related to the infiltration rate, these reweighting schemes would shift the estimated coefficient. We find stable estimates across weighting schemes, suggesting that our main results are not driven by differential data coverage across hours, households, or their interaction (*SI Appendix*, Fig. S4).

Outdoor $PM_{2.5}$: Our hardware partner, Nafas, maintains a dense network of low-cost Airly outdoor sensors across the city, each equipped with a Plantower PMS5003 optical light-scattering sensor. Accuracy and network performance follow the manufacturer's factory calibration and Nafas's routine QA/QC procedures. This sensor has been broadly deployed and is one of the most extensively studied low-cost sensor. It has been shown to have relatively high accuracy against reference instruments with high hourly correlations (R^2 typically between 0.80 and 0.95) (30, 31), demonstrating strong temporal tracking despite systematic bias in absolute mass concentrations, especially at high levels of humidity. While there is no published side-by-side evaluation of the PMS5003 and PM2008M, systematic bias would not qualitatively affect the estimated infiltration factors so long as calibration slopes are stable because our analysis relies primarily on temporal co-variation and infiltration estimates rather than absolute mass differences between indoor and outdoor sensors.

For each household, we calculate a weighted average of air quality readings from the three nearest outdoor sensors, assigning weights inversely proportional to the distance to each sensor. The 63 outdoor sensors used in this study also capture temperature and humidity data, which we use as controls. The outdoor low-cost sensors were installed prior to our study and were not randomly distributed across space, which could introduce non-randomly distributed measurement error. We test for selection in the location of outdoor sensors by regressing a household's distance to the closest outdoor sensor on recorded indoor $PM_{2.5}$, recorded outdoor $PM_{2.5}$ of this sensor, household income, house size, whether the household owns an AC, household head employment status, and household head education level. We do not find evidence of selection bias on observables, as shown in the *SI Appendix*, Table S4.

Natural ventilation behaviors: To complement our main survey efforts, we deployed a short WhatsApp survey to participating households on three separate occasions during the study period. The survey asked respondents whether the windows and doors in the room housing the indoor monitor had been open or closed during the past hour.

B. Infiltration Factor. The average indoor infiltration factor is defined as the increase in indoor $PM_{2.5}$ concentration per unit increase in local ambient outdoor $PM_{2.5}$ concentration. To estimate this factor, we used a regression model at the monitor-hour level. For each household i in hour t, the infiltration factor (β) was estimated using the following specification:

$$\begin{aligned} PM_{it}^{In} = \alpha &+ \beta PM_{it}^{Out} \\ &+ \pi_1 \mathbf{1}[\text{Smoking}]_i \\ &+ \pi_2 \mathbf{1}[\text{Waste burning}]_i + \pi_3 \mathbf{1}[\text{Kitchen}]_i \\ &+ \pi_4 \mathbf{1}[\text{CookingHour}]_t + \pi_5 \text{DistPrimaryRoad}_i \\ &+ \psi T_{it} + \gamma H_{it} + \mu_h + \mu_w + \varepsilon_{it} \end{aligned} \quad [1]$$

where PM_{it}^{In} and PM_{it}^{Out} denote indoor and outdoor $PM_{2.5}$ concentrations for household i at hour t, respectively. $\mathbf{1}$[Smoking] indicates whether the respondent reported smoking inside the home in the 24 hours preceding the survey. $\mathbf{1}$[Waste burning] is a categorical indicator for self-reported, waste burning activity in the neighborhood in the seven days preceding the survey, distinguishing no burning events, one to two events, and more than two events. $\mathbf{1}$[Kitchen] indicates whether the monitor was located in or adjacent to the kitchen. $\mathbf{1}$[Cooking hour] is a binary indicator equal to one if, according to a 24-hour time-use recall survey, the household reported cooking during the hour of day corresponding to t. DistPrimaryRoad is the household's distance to the nearest primary road, defined as a highway or the second-highest tier of road in the OpenStreetMap classification. T_{it} and H_{it} are contemporaneous outdoor temperature and relative humidity, respectively. μ_h denotes hour-of-day fixed effects, capturing diurnal patterns in both ambient pollution and household activity, and μ_w denotes week-of-sample fixed effects, capturing longer-run temporal trends and atmospheric conditions.

The estimated β for this main specification and alternative ones is presented in Fig. 2:

1. Main specification: includes household-specific controls that account for non-ambient sources, controls for average variation in $PM_{2.5}$ over the course of the day (hour-of-day fixed effects, μ_h) and average weekly trends (week-of-sample fixed effects, μ_w).

2. Controls for average variation in $PM_{2.5}$ over the course of the day (hour-of-day fixed effects, μ_h), average weekly trends (week-of-sample fixed effects, μ_w), as well as time-invariant differences across households (μ_i).

3. Controls for average weekly trends (week-of-sample fixed effects, μ_w) and for household-specific average variation in $PM_{2.5}$ at different times of the day (household-hour-of-day fixed effects, μ_{ih}).

4. Specification 1 but restricts the sample to households located within 2 km of a low-cost outdoor sensor to assess the potential impact of measurement error in outdoor $PM_{2.5}$ concentrations on infiltration estimates.

5. Specification 1, but pollution data is aggregated into 8 hour means.

6. Specification 1, but pollution data is aggregated into 24 hour means.

The estimated infiltration factor (β) remains highly stable across all specifications. We chose specification 1, which includes as our main specification as it enables us to examine heterogeneity by household-level characteristics, which would

DRAFT

not be possible in the presence of household fixed-effects (μ_i).

We further investigate lagged effects of outdoor $PM_{2.5}$ with the following specification:

$$\begin{aligned} PM_{it}^{In} = \alpha &+ \sum_{k=0}^{11} \beta_k PM_{it}^{Out} \\ &+ \pi_1 \mathbf{1}[\text{Smoking}]_i \\ &+ \pi_2 \mathbf{1}[\text{Waste burning}]_i + \pi_3 \mathbf{1}[\text{Kitchen}]_i \\ &+ \pi_4 \mathbf{1}[\text{CookingHour}]_t + \pi_5 \text{DistPrimaryRoad}_i \\ &+ \psi T_{it} + \gamma H_{it} + \mu_h + \mu_w + \varepsilon_{it} \end{aligned} \quad [2]$$

In *SI Appendix*, Fig. S5, we present the results when including 11 lags (outdoor $PM_{2.5}$ at each of the previous 11 hours) and ensure that the results are robust to the inclusion of additional lags. From this regression, we derive an estimate of outdoor-indoor infiltration by calculating the cumulative effect of a 1 $\mu g/m^3$ increase in outdoor concentration on indoor concentration:

$$\beta' = \sum_{k=0}^{1} 1\beta_k \quad [3]$$

Infiltration factors derived from this lagged regression are very similar to our main estimates, consistent with most of the infiltration happening within the first hour, as shown in Fig. 2.

To examine heterogeneity in infiltration factors across household characteristics X_i, we run the following interacted specification:

$$\begin{aligned} PM_{it}^{In} = \alpha &+ \beta_1 PM_{it}^{Out} + \beta_2 X_i + \beta_3 (PM_{it}^{Out} \times X_i) \\ &+ \pi_1 \mathbf{1}[\text{Smoking}]_i \\ &+ \pi_2 \mathbf{1}[\text{Waste burning}]_i + \pi_3 \mathbf{1}[\text{Kitchen}]_i \\ &+ \pi_4 \mathbf{1}[\text{CookingHour}]_t + \pi_5 \text{DistPrimaryRoad}_i \\ &+ \psi T_{it} + \gamma H_{it} + \mu_h + \mu_w + \varepsilon_{it} \end{aligned} \quad [4]$$

C. Decomposition of Indoor $PM_{2.5}$. To decompose indoor $PM_{2.5}$ concentrations by source for Fig. 3, we use the coefficients from Equation 1. We quantify each source's mean contribution to indoor $PM_{2.5}$ by multiplying the estimated coefficient by the sample mean of the corresponding variable and expressing the product as a share of mean indoor $PM_{2.5}$.

We further assess source importance by decomposing the R^2 of Equation 1 across predictors using the Lindeman-Merenda-Gold (LMG) method (*SI Appendix*, Tables S2 and S3) (32, 33).

To quantify how income explains differences in indoor PM2.5 exposure, we estimate a single regression of indoor PM2.5 on all source variables interacted with income quartile, controlling for outdoor temperature, humidity, and hour-of-day and week fixed effects. This allows the coefficients on outdoor ambient PM2.5 (the infiltration factor) and each hyperlocal source to vary by income quartile. We then decompose mean indoor PM2.5 for each quartile into three components: the outdoor ambient contribution, calculated by multiplying the quartile-specific infiltration rate by the quartile-specific mean outdoor PM2.5; the smoking contribution, calculated analogously using quartile-specific smoking coefficients and prevalence; and a residual hyperlocal component, computed as the difference between observed mean indoor PM2.5 and the sum of the outdoor and smoking contributions.

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DRAFT

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DRAFT